Civil Engineering

Trigonometry, Bearings and Complex Numbers

Assignment 1

Instructions: Answer all questions. Show formula, substitution, calculation, units and technical interpretation.

Section A: Formula and Concept Questions

1. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$a^2+b^2=c^2$$

2. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\sin\theta=\frac{O}{H}$$

3. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$c^2=a^2+b^2-2ab\cos C$$

4. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

5. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$z=a+bi$$

Section B: Applied Problems

6. Solve a civil engineering calculation problem involving heights. Show all working and write one technical interpretation. (10 marks)

7. Solve a civil engineering calculation problem involving slopes. Show all working and write one technical interpretation. (10 marks)

8. Solve a civil engineering calculation problem involving triangulation. Show all working and write one technical interpretation. (10 marks)

9. Solve a civil engineering calculation problem involving bearings. Show all working and write one technical interpretation. (10 marks)

10. Solve a civil engineering calculation problem involving polar representation. Show all working and write one technical interpretation. (10 marks)

11. Solve a civil engineering calculation problem involving heights. Show all working and write one technical interpretation. (10 marks)

12. Solve a civil engineering calculation problem involving slopes. Show all working and write one technical interpretation. (10 marks)

13. Solve a civil engineering calculation problem involving triangulation. Show all working and write one technical interpretation. (10 marks)

14. Solve a civil engineering calculation problem involving bearings. Show all working and write one technical interpretation. (10 marks)

15. Solve a civil engineering calculation problem involving polar representation. Show all working and write one technical interpretation. (10 marks)

16. Solve a civil engineering calculation problem involving heights. Show all working and write one technical interpretation. (10 marks)

17. Solve a civil engineering calculation problem involving slopes. Show all working and write one technical interpretation. (10 marks)

18. Solve a civil engineering calculation problem involving triangulation. Show all working and write one technical interpretation. (10 marks)

19. Solve a civil engineering calculation problem involving bearings. Show all working and write one technical interpretation. (10 marks)

20. Solve a civil engineering calculation problem involving polar representation. Show all working and write one technical interpretation. (10 marks)

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.